

This assignment will not be graded and is only for practice.

Level 1

1) Choose the set of correct options.

1 point

- ☐ If v_1 and v_2 are in the kernel of a linear transformation $T : V \rightarrow W$, where V and W are two vector spaces, then $v_1 + v_2$ will also be in the kernel of T .
- ☐ Kernel of a linear transformation $T : V \rightarrow W$ is a vector subspace of V , where V and W are two vector spaces.
- ☐ If w_1 and w_2 are in the image of a linear transformation $T : V \rightarrow W$, where V and W are two vector spaces, then $w_1 + w_2$ will also be in the image of T .
- ☐ Image of a linear transformation $T : V \rightarrow W$ is a vector subspace of W , where V and W are two vector spaces.

No, the answer is incorrect.

Score: 0

Accepted Answers:

If v_1 and v_2 are in the kernel of a linear transformation $T : V \rightarrow W$, where V and W are two vector spaces, then $v_1 + v_2$ will also be in the kernel of T .

Kernel of a linear transformation $T : V \rightarrow W$ is a vector subspace of V , where V and W are two vector spaces.

If w_1 and w_2 are in the image of a linear transformation $T : V \rightarrow W$, where V and W are two vector spaces, then $w_1 + w_2$ will also be in the image of T .

Image of a linear transformation $T : V \rightarrow W$ is a vector subspace of W , where V and W are two vector spaces.

Consider the following linear transformation:

$$T : \mathbb{R}^3 \rightarrow \mathbb{R}^2$$

$$T(x, y, z) = (2x + 3z, 4y + z)$$

Answer questions 2,3,4 and 5, using the information given above.

2) Which of the following matrices corresponds to the given linear transformation T **1 point** with respect to the standard ordered basis for $\mathbb{R}^3$ and the standard ordered basis for $\mathbb{R}^2$?

☐ $\begin{bmatrix} 2 & 3 & 0 \\ 4 & 1 & 0 \end{bmatrix}$

☐ $\begin{bmatrix} 2 & 0 \\ 0 & 4 \\ 3 & 1 \end{bmatrix}$

☐ $\begin{bmatrix} 2 & 0 & 3 \\ 0 & 4 & 1 \end{bmatrix}$

☐ $\begin{bmatrix} 2 & 4 \\ 3 & 1 \\ 0 & 0 \end{bmatrix}$

No, the answer is incorrect.

Score: 0

Accepted Answers:

$$\begin{bmatrix} 2 & 0 & 3 \\ 0 & 4 & 1 \end{bmatrix}$$

3) Which of the following represents a basis of the kernel of T ?

1 point

☐ $\{(-\frac{3}{2}, 0, 1), (0, -\frac{1}{4}, 1)\}$

☐ $\{(-\frac{3}{2}, -\frac{1}{4}, 1)\}$

☐ $\{(-\frac{3}{2}, -\frac{1}{4}, 2)\}$

☐ $\{(2, 0, 3), (0, 4, 1)\}$

No, the answer is incorrect.

Score: 0

Accepted Answers:

$$\{(-\frac{3}{2}, -\frac{1}{4}, 1)\}$$

4) What will be the dimension of the subspace $Im(T)$?

No, the answer is incorrect.

Score: 0

Accepted Answers:

(Type: Numeric) 2

1 point

5) Choose the correct option.

1 point

- ☐ T is an isomorphism.
- ☐ T is one to one but not onto.
- ☐ T is onto but not one to one.
- ☐ T is neither one to one, nor onto.

No, the answer is incorrect.

Score: 0

Accepted Answers:

T is onto but not one to one.

6) Choose the set of correct options.

1 point

- ☐ There exists a linear transformation $T : \mathbb{R}^2 \rightarrow \mathbb{R}^2$ whose range is contained in the circle $x^2 + y^2 = 1$.
- ☐ There exists a linear transformation $T : \mathbb{R}^2 \rightarrow \mathbb{R}^2$ whose range equals the line $x = y$.
- ☐ There exists a linear transformation $T : \mathbb{R}^2 \rightarrow \mathbb{R}^2$ whose range is contained in the set $S = \{(x, y) \mid x > 0\}$.
- ☐ There exists a linear transformation $T : \mathbb{R}^2 \rightarrow \mathbb{R}^2$ whose range equals the line $y = \pi x$.

No, the answer is incorrect.

Score: 0

Accepted Answers:

There exists a linear transformation $T : \mathbb{R}^2 \rightarrow \mathbb{R}^2$ whose range equals the line $x = y$.

There exists a linear transformation $T : \mathbb{R}^2 \rightarrow \mathbb{R}^2$ whose range equals the line $y = \pi x$.

Level 2

7) Which option represents the kernel and image of the following linear transformation?

1 point

$$T : \mathbb{R}^2 \rightarrow \mathbb{R}^2$$

$$T(x, y) = (x, 0)$$

- ☐ $\ker(T) = \text{Span}\{(1, 0)\}, \text{Im}(T) = \text{Span}\{(1, 0)\}$
- ☐ $\ker(T) = \text{Span}\{(1, 0)\}, \text{Im}(T) = \text{Span}\{(0, 1)\}$
- ☐ $\ker(T) = \text{Span}\{(0, 1)\}, \text{Im}(T) = \text{Span}\{(1, 0)\}$
- ☐ $\ker(T) = \text{Span}\{(0, 1)\}, \text{Im}(T) = \text{Span}\{(0, 1)\}$

No, the answer is incorrect.

Score: 0

Accepted Answers:

$$\ker(T) = \text{Span}\{(0, 1)\}, \text{Im}(T) = \text{Span}\{(1, 0)\}$$

8) Which of the following linear transformations is an isomorphism between the vector spaces $V = \{(x, y, z) \mid x = y - z, \text{ and } x, y, z \in \mathbb{R}\} \subset \mathbb{R}^3$ and $W = \{(x, y, z) \mid x = y, \text{ and } x, y, z \in \mathbb{R}\} \subset \mathbb{R}^3$? **1 point**

- ☐ $T : V \rightarrow W$, such that $T(x, y, z) = (y, y, 0)$
- ☐ $T : V \rightarrow W$, such that $T(x, y, z) = (y, y, z)$
- ☐ $T : V \rightarrow W$, such that $T(x, y, z) = (x, y, z)$
- ☐ There does not exist any isomorphism between V and W .

No, the answer is incorrect.

Score: 0

Accepted Answers:

$$T : V \rightarrow W, \text{ such that } T(x, y, z) = (y, y, z)$$

9) Which of the following linear transformations is an isomorphism between the vector **1 point** spaces $V = \{(x, y, z) \mid x = y - z = 0, \text{ and } x, y, z \in \mathbb{R}\} \subset \mathbb{R}^3$ and $W = \{(x, y, z) \mid x = y = 0, \text{ and } x, y, z \in \mathbb{R}\} \subset \mathbb{R}^3$?

- ☐ $T : V \rightarrow W$, such that $T(x, y, z) = (x, 0, y)$.
- ☐ $T : V \rightarrow W$, such that $T(x, y, z) = (0, y, z)$.
- ☐ $T : V \rightarrow W$, such that $T(x, y, z) = (0, 0, x)$.
- ☐ There does not exist any isomorphism between V and W .

No, the answer is incorrect.

Score: 0

Accepted Answers:

$T : V \rightarrow W$, such that $T(x, y, z) = (x, 0, y)$.

10) Consider the linear transformation $T : \mathbb{R}^3 \rightarrow \mathbb{R}^3$ such that $T(x, y, z) = (x - y + z, 2x + y - z, -x - 2y + 2z)$. Which of the following options are true about T ? **1 point**

- ☐ Range of T is $\{(x, y, z) \in \mathbb{R}^3 \mid x = y + z\}$.
- ☐ Range of T is $\{(x, y, z) \in \mathbb{R}^3 \mid x = 0, y + z = 0\}$.
- ☐ Rank of T is 2
- ☐ Rank of T is 1

No, the answer is incorrect.

Score: 0

Accepted Answers:

Range of T is $\{(x, y, z) \in \mathbb{R}^3 \mid x = y + z\}$.

Rank of T is 2

Your score is: 0/10